

5. (a) Newton's second law of motion is sometimes expressed as $\Sigma F = ma$. Explain the term ΣF , giving an example to illustrate your answer. [2]

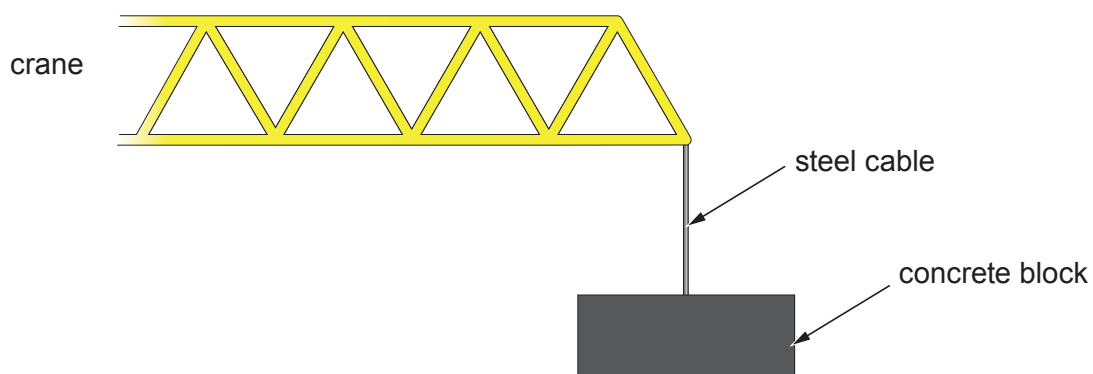
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- (b) A crane uses a steel cable to lift heavy objects on a building site. The crane operator is told that the maximum safe lifting force of the crane is 16 000 N.



- (i) A concrete block of mass 1 500 kg is attached to the crane. Calculate the maximum safe upward acceleration of the block. [3]

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- (ii) The crane operator assumes that the crane is safe to lift any load up to 16 000 N. Discuss whether or not he is correct. [2]

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- (c) The steel cable has a cross-sectional area of $2.0 \times 10^{-3} \text{ m}^2$ and a Young modulus of $2.0 \times 10^{11} \text{ N m}^{-2}$. As the concrete block moves **upwards** the tension in the cable changes depending on whether the block is accelerating, decelerating or moving at constant speed. At one point in its motion the strain in the cable is 3.2×10^{-5} . Describe the motion of the block at this point. Explain your answer. [4]

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